

HS-401: Professional Values and Ethics

LECTURE NO 2: *The Responsibilities of Engineers*

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1.1 Introduction

This Lecture first investigates what exactly responsibility is, distinguishing between passive responsibility and active responsibility. Then we will discuss the position of engineers vis-à-vis managers, then, the wider context of technological development, and examine the consequences for the responsibility of engineers of this wider context.

1.2 Responsibility

- Whenever something goes wrong or there is a disaster then the question who is responsible for it often quickly arises. Here responsibility means in the first place being held accountable for your actions and for the effects of your actions. The making of choices, the taking of decisions but also failing to act are all things that we regard as types of actions. There are different kinds of responsibility that can be distinguished.
 - *Active Responsibility*- Active responsibility is responsibility before something has happened.
 - *Passive Responsibility*- Passive responsibility is applicable after something (undesirable) has happened.
- Responsibility (both active and passive) is often linked to the role that you have in a particular situation.

1.2 Responsibility (continued...)

- You often have to fulfil a number of roles simultaneously such as those of friend, parent, citizen, employee, engineer, expert, and colleague. In a role you have a relationship with others, for instance, as an employee you have a relationship with your employer, as an expert you have a relationship with your customers and as a colleague you have relationships with other colleagues.
- Each role brings with it certain responsibilities. A parent, for example, is expected to care for his child. In the role of employee it is expected that you will execute your job properly.
- An engineer is expected to carry out his work in a competent way.
- Roles and their accompanying responsibilities can be formally laid down, for instance legally, in a contract or in professional or corporate codes of conduct.
 - **Role responsibility-** The responsibility that is based on the role one has or plays in a certain situation.
 - **Moral responsibility-** Although roles define responsibilities, moral responsibility is not confined to the roles one plays in a situation. Rather it is based on moral obligations, moral norms and moral duties.

1.2 Responsibility (continued...)

- *Professional responsibility*- is the responsibility that is based on your **role** as a professional engineer as far as it stays within the limits of what is **morally** allowed. Professional responsibilities are not just passive but they also contain an active component.

1.3 Passive Responsibility

- Typical for passive responsibility is that the person who is held responsible must be able to provide an account why he/she followed a particular course of action and why he/she made certain decisions.
- *Passive responsibility*- Backward-looking responsibility, relevant after something undesirable occurred; specific forms are accountability, blameworthiness, and liability.
 - *Accountability*- Backward-looking responsibility in the sense of being held to account for, or justify one's actions towards others.
 - *Blameworthiness*- Blameworthiness means that it is proper to blame someone for his/her actions or the consequences of those actions. Usually, four conditions need to apply wrong-doing, causal contribution, foreseeability, and freedom.

1.3 Passive Responsibility (continued...)

- *Wrong doing*- Whenever one blames a person or institution one usually maintains that in carrying out a certain action the individual or the institution in question has violated a norm or did something wrong. This can be a norm that is laid down in the law or that is common in the organization
- *Causal contribution*- A second criterion is that the person who is held responsible must have made a causal contribution to the consequences for which he or she is held responsible. Two things are to be kept in mind when judging whether someone made a causal contribution to a certain consequence. First, not only an action, but also a failure to act may often be considered a causal contribution. Second, a causal contribution is usually not a sufficient condition for the occurrence of the consequence under consideration. Often, a range of causal contributions will have to be present for the consequence to occur. A causal contribution will often be a necessary ingredient in the actual chain of events that led to the consequence, that is, without the causal contribution the consequence would not have occurred.

1.3 Passive Responsibility (continued...)

- *Foreseeability*- A person who is held responsible for something must have been able to know the consequences of his or her actions.
- *Freedom of action*- Finally, the one who is held responsible must have had freedom of action, that is, he or she must not have acted under compulsion.

1.4 Active Responsibility and the Ideals of Engineers

- Responsibility is also something that comes into play beforehand, if nothing has yet gone wrong or if there is the chance to realize something good. We will refer to this as *active responsibility*. If someone is actively responsible for something he/she is expected to act in such away that undesired consequences are avoided as much as possible and so that positive consequences are realized.
- One way in which the active responsibility of engineers can be understood is by looking at the *ideals* of engineers.
- **Ideals** are ideas or strivings which are particularly motivating and inspiring for the person having them, and which aim at achieving an optimum or maximum.
- **Professional ideals-** Ideals that are closely allied to a profession or can only be aspired to by carrying out the profession.

1.4 Active Responsibility and the Ideals of Engineers (continued...)

Three different professional ideals of engineers are discussed below:

1.4.1 *Technological enthusiasm-*

The ideal of wanting to develop new technological possibilities and taking up technological challenges.

1.4.2 Effectiveness and Efficiency- Engineers tend to strive for effectiveness and efficiency. *Effectiveness* can be defined as the extent to which an established goal is achieved; *efficiency* as the ratio between the goal achieved and the effort required. The drive to strive towards effectiveness and efficiency is an attractive ideal for engineers because it is – apparently – so neutral and objective.

1.4.3 Human Welfare- A third ideal of engineers is that of contributing to or augmenting human welfare. The professional code of the American Society of Mechanical Engineering (ASME) and of the American Society of Civil Engineers (ASCE) states that “engineers shall use their knowledge and skill for the enhancement of human welfare.” This also includes values such as health, the environment, and sustainability.

1.5 Engineers versus Managers

- Engineers are often salaried employees and they are usually hierarchically below managers. Just as with other professionals this can lead to situations of conflict, because on the one hand, they have a responsibility to the company in which they work and, on the other hand, a professional responsibility as engineers, including — as we have seen — a responsibility for human welfare.
- We will discuss three models of dealing with this tension and the potential conflict between engineers and managers: *separatism*, *technocracy*, and *whistle-blowing*. These three models are positions that engineers can adopt versus managers in specific situations.

1.5 Engineers versus Managers (continued...)

1.5.1 Separatism- The notion that scientists and engineers should apply the technical inputs, but appropriate management and political organs should make the value decisions.

1.5.2 Technocracy- Frederick Taylor, an American mechanical engineer, proposed that engineers should takeover the role of *managers* in the governance of companies and that of *politicians* in the governance of society. This proposal would lead to the establishment of a **technocracy**, that is, government by experts. Accordingly, the role of engineers would be that of technocrats who, on the basis of technological insight, do what they consider best for a company or for society.

1.5.3 Whistle-Blowing- The term whistle-blowing is used if an employee discloses certain abuses in a company in which he/she is employed without the consent of his/her superiors and in order to remedy these abuses and/or to warn the public about these abuses.

1.6 The Social Context of Technological Development

- Engineers are not the only ones who are responsible for the development and consequences of technology. Apart from managers and engineers there are other actors that influence the direction taken by technological development and the relevant social consequences.
- We use the term *actor* here for any person or group that can make a decision how to act and that can act on that decision.
- A company is an actor because it usually has a board of directors that can make decisions on behalf of that company and is able to effectuate those decisions. A variety of actors can be distinguished that usually play a role in technological development:
 - *Developers and producers of technology.* This includes engineering companies, industrial laboratories, consulting firms, universities and research centers, all of which usually employ scientists and engineers.

1.6 The Social Context of Technological Development (continued...)

- *Users* who use the technology and formulate certain requirements for the functioning of the technology. The users of technologies are a very diverse group, including both companies and citizens (consumers).
- *Regulators* such as the government, who formulate rules or regulations that engineering products have to meet such as rulings concerning health and safety, but also rulings linked to relations between competitors.
- Also other actors may be involved in technological development including, for example, professional associations, educational institutes, interest groups etc (see Figure 1 next slide). All these actors have certain *interests*,— things actors strive for because they are beneficial or advantageous for them. The interests of the various actors will often conflict, so that there is no agreement on the desirable direction of technological development.

1.6 The Social Context of Technological Development (continued...)

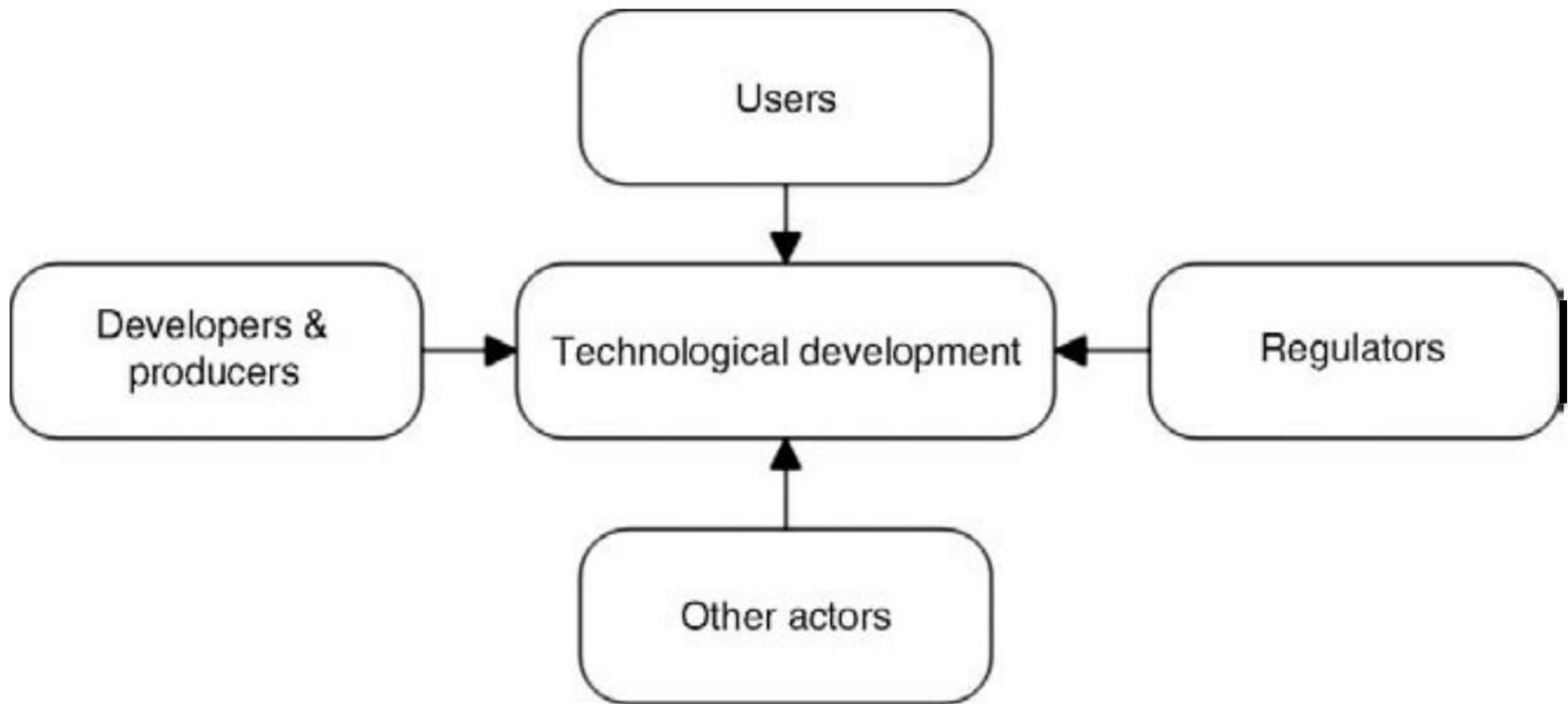


Figure 1: Technological development map of actors.

1.6 The Social Context of Technological Development (continued...)

- Also In addition to actors that influence the direction of technological development we distinguish stakeholders. *Stakeholders* are actors that have an interest (“a stake”) in the development of a technology, but who cannot necessarily influence the direction of technological development. An example is people living in the vicinity of a planned construction site for a nuclear plant. Obviously these people have an interest in what type of reactor is built and how safe it is but they may not be able to influence the technology developed.
- Technological development is an unpredictable process. In the course of time, a variety of methods and approaches have been developed to deal with this unpredictable character of technology development. This is done by a discipline known as *Technology Assessment (TA)*. TA is a Systematic method for exploring future technology developments and assessing their potential societal consequences.

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